



Autumn Examinations 2017 / 2018

Exam Code(s) 1MDM1, 1MF1, 1SD1, 1SD3
Exam(s) Masters in Software Design & Development
Higher Diploma in Applied Science (Software Design & Development)
Masters in Digital Media

Module Code(s) CT870
Module(s) Internet Programming

Paper No. 1

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Instructions: Answer any 3 questions. All questions carry equal marks.

Duration 2hrs
No. of Pages 3 (Including cover page)
Department(s) Information Technology

Requirements None

Release to Library: Yes ☐ No ☒

Question 1:

- a) Describe 5 different tags that can be used on a HTML page. Illustrate each one with an example. (4 marks)
- b) Write HTML code that creates an Unordered List of 12 elements representing the 12 months of the year (January, February, March, etc.). (4 marks)
- c) Write HTML that creates a table that represents a sports league. There should be 6 columns for Name, Played, Won, Drawn, Lost and Points. Include 3 rows of dummy data. (6 marks)
- d) Write HTML for a login form which contains a dropdown list (options for Male/Female), two text boxes (one for username and one for password), a checkbox for “remember me”, a submit and a reset button. (6 marks)

20 Marks

Question 2:

- a) Explain how the grid system in Bootstrap works. In your answer discuss the purpose of rows and columns and describe how the grid facilitates responsive design. (5 marks)
- b) Provide the HTML and corresponding Bootstrap classes to split a row into two separate sections, each sized 6 columns wide on a large device and medium device. In the case of smaller devices, the columns should stack on top of each other. (5 marks)
- c) How are fonts represented in CSS? Give an example to show your understanding. (5 marks)
- d) In CSS how can a single style be applied to all elements? (5 marks)

20 Marks

Question 3:

- a) Using JavaScript, iterate through an array of objects (called myArray). The array contains a number of car objects which take the following form ({**make**: “Ford”, **model**: “Focus”, **age**:2} , ... , {**make**: “Toyota”, **model**: “Yaris”, **age**:5}). The ages of each car in the array were incorrectly specified originally and all must be increased by 2 in order to rectify the problem. Provide code to do this. (8 marks)
- b) Describe 2 methods that can be used to perform client-side validation. (4 marks)
- c) Write JavaScript code that upon button click reads 3 numbers from 3 text boxes and doubles each number before saving it in the text box it came from. (8 marks)

20 Marks

PTO

Question 4:

You have been asked to build a backend, which comprises a collection of RESTful APIs for an Instagram style application using NodeJS (Express). Each image shared by the users of the application should contain the GPS coordinates of the location of where the image was taken (Long. and Lat.), the upload time and date, the number of likes and the number of re-shares.

- a) Provide a sample JSON document which contains data for a single image shared on the platform. (4 marks)
- b) Write a MongoDB or Mongoose query which returns all images posted, sorted in ascending order by upload time. (5 marks)
- c) Define a server-side API route accessible at '/uploadImages' which handles POST requests. Assume that that MIME type is application/json and the payload is processed by the Express middleware and made available as an object in the request body. Either using the object modelling tool Mongoose or a traditional MongoDB query, provide code to save the data in the request body to the MongoDB database. If using Mongoose, you do not need to specify the schema, you may assume a schema named Image is already defined. (7 marks)
- d) Define a server-side API to delete documents from the collection by parsing the request parameters, extracting the id and removing the document found. (4 marks)

20 Marks